Vector Valued Functions

section 12.1
9-11-26

A) Curves in Space

$Y(t) = \langle f(t), g(t) \rangle$

$y(t) = \langle f(t), g(t), h(t) \rangle$

Ex) $y = x^2 - 3$

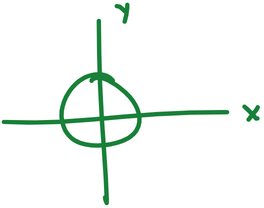
$\langle t, t^2 - 3 \rangle$ $\quad x = t$

$\langle \sqrt{t+3}, t \rangle$ $\quad y = t$

$\langle t^2, t^4 - 3 \rangle$ $\quad x = t^2$

$\langle \sqrt{t}, t^{-3} \rangle \quad x = \sqrt{t}$

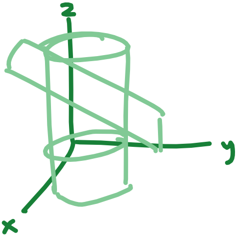
Ex) $x^2+y^2=4$



$\langle 2\cos t, 2\sin t \rangle$

$\langle 2\sin t, 2\cos t \rangle$

Ex) $x^2+y^2=1 \quad y+z=2$



$\langle \cos t, \sin t, 2-\sin t \rangle$

Ex) $x^2+y^2-4z^2=1$ and $y=x^2$

state curve of intersection of the spaces as a vector-valued function

$$\left\langle t, t^2, \frac{\sqrt{t^2+t^4-1}}{2} \right\rangle$$

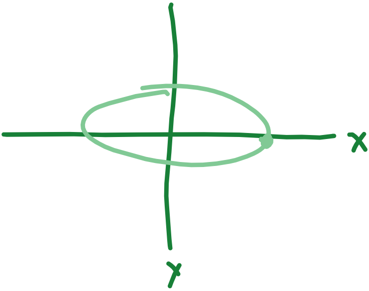
$x=t$

$y=t^2 \qquad t^2+t^4-4z^2=1$

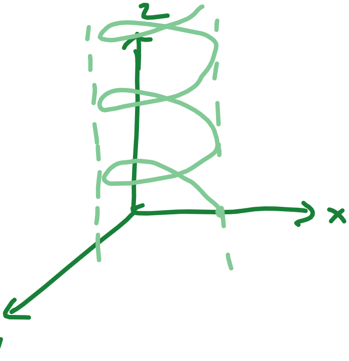
$t^2+t^4-1 = 4z^2$

B) Sketch the Curve

ex) $r(\theta) = \langle 3\cos\theta, \sin\theta \rangle$



ex) $r(t) = \langle 3\cos t, \sin t, t \rangle$

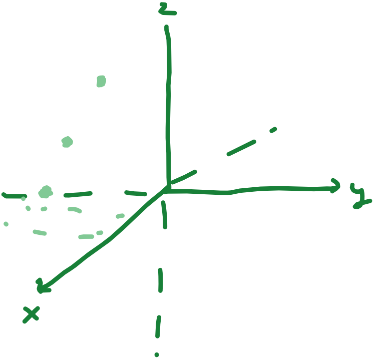


ex) $r(t) = \langle t, 2t-8, 3t \rangle$

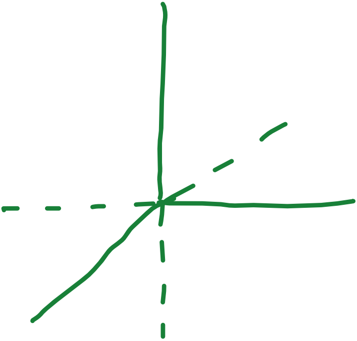
$r(0) = (0, -5, 0)$

$r(1) = (1, -3, 3)$

$r(2) = (2, -1, 6)$



ex) $r(t) = \langle t, t^2, t^3 \rangle$



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Not sure why #s repeat...

19) VI   20) II   21) IV

22) I   23) V   24) III

21) II   22) VI   23) V

24) I   25) IV   26) III

19) B   20) C   21) D   22) A

45) $\langle t, t+5 \rangle$

46) $\langle \frac{t}{2}, \frac{t}{3} + \frac{5}{3} \rangle$

47) $\langle t+2, t^2 \rangle$

48) $\langle \sqrt{t}, 4-t \rangle$

49) $\langle 5\cos t, 5\sin t \rangle$

50) $\langle 2\cos t+2, 2\sin t \rangle$

51) $\langle 4\sec t, 2\tan t \rangle$

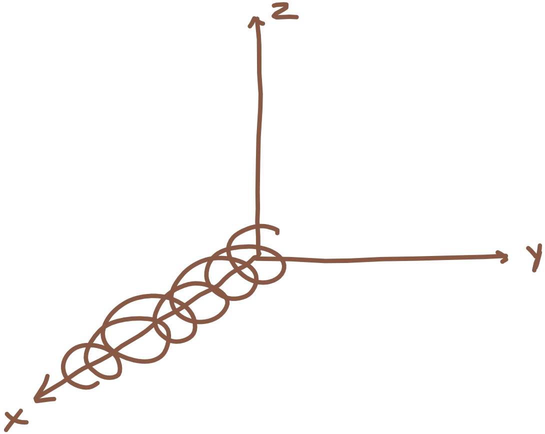
52) $\langle 3\cos t, 4\sin t \rangle$

OR

$\langle 3 \sin t \ , \ 4 \cos t \rangle$

34) $r(t) = t(i) + 3\cos t(j) + 3\sin t(k)$

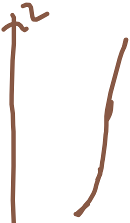
$\langle t, 3\cos t, 3\sin t \rangle$

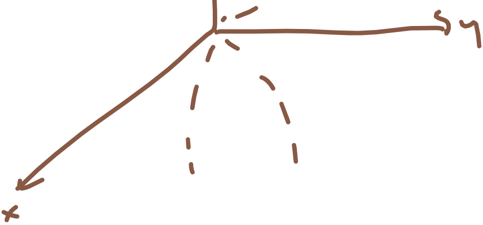


37) $r(t) = \langle t, t^2, \frac{2}{3} t^3 \rangle$

$r(-1) = \langle -1, 1, -\frac{2}{3} \rangle$

$r(0) = \langle 0, 0, 0 \rangle$



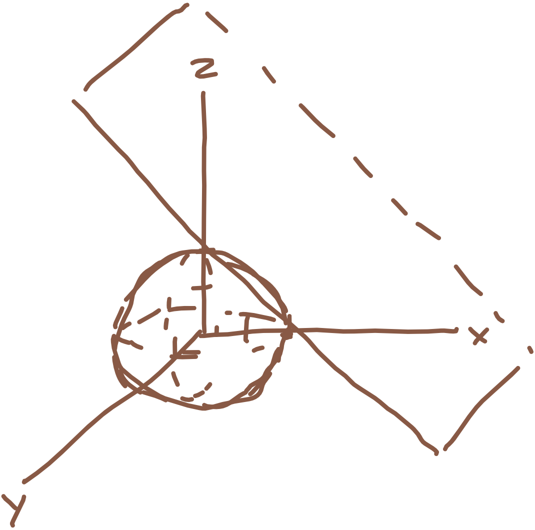


55) $x^2+y^2=4,\ z=x^2$ $\qquad x=2\sin t$

$\langle 2\sin t, 2\cos t, 4\sin^2 t$

57) $x^2+y^2+z^2=4,\ x+z=2$ $\qquad x=1+\sin t$

$\langle 1+\sin t, \sqrt{2}\cos t, 1-\sin t\rangle$



C) Limits and Continuity

1) $\lim_{t\to\infty} r(t) = \langle \lim_{t\to\infty} f(t), \lim_{t\to\infty} g(t), \lim_{t\to\infty} h(t) \rangle$

2) r(t) is cont. at a if . ...

ex)

$r(t) = \langle t, 3, 9-t^2 \rangle$

is r(t) cont. at t=0?

$r(0) = \langle 0, 3, 9 \rangle$

$\lim_{t\to\infty} r(t) = \langle \lim_{t\to\infty} t, \lim_{t\to\infty} 3, \lim_{t\to\infty} 9-t^2 \rangle$

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5) $r(t) = F(t) + G(t)$

$F(t) = (\cos t)i - (\sin t)j + (\sqrt{t})k$

$G(t) = (\cos t)i + (\sin t)j$

domain:

6) $(0, \infty)$

7) $\langle \cos^2 t, \sin t \cos t, \sin^2 t \rangle$

8) $\langle -t^2 + 2t - \frac{t}{t+1}, -(t^4 + 2t^3), \frac{t^3}{t+1} - t^4 - 2t^3 \rangle$

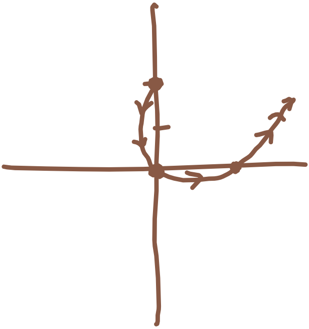
26) $r(t) = (t^2 + t)i + (t^2 - t)j$

$r(-1) = (0, 2)$

$r(0) = (0, 0)$

$r(1) = (2, 0)$

$r(2) = (6, 2)$



64) $6i + \frac{2}{3}j + \frac{1}{2}k$

65) $(0, 0, 0)$

68) (0, 0, 0)

87) x' : t^2-3t+4

$(t-4)(t+3)$

$\underline{y'}$ $t^2-9t+20$

$(t-5)(t-4)$

$\underline{z'}$ t^2-5t+4

$(t-1)(t-4)$

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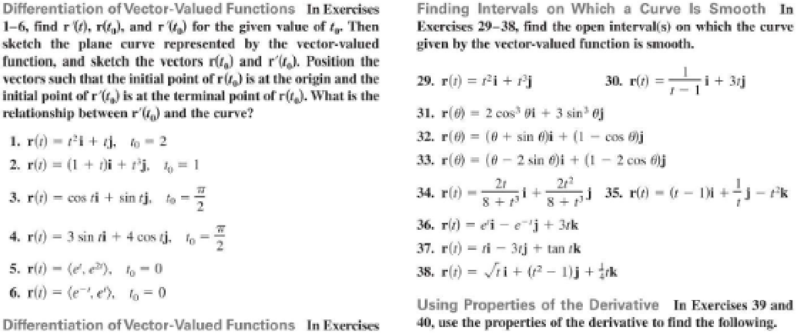

Differentiation of Vector-Valued Functions In Exercises 1–6, find $\mathbf{r}'(t)$, $\mathbf{r}(t_0)$, and $\mathbf{r}'(t_0)$ for the given value of t_0. Then sketch the plane curve represented by the vector-valued function, and sketch the vectors $\mathbf{r}(t_0)$ and $\mathbf{r}'(t_0)$. Position the vectors such that the initial point of $\mathbf{r}(t_0)$ is at the origin and the initial point of $\mathbf{r}'(t_0)$ is at the terminal point of $\mathbf{r}(t_0)$. What is the relationship between $\mathbf{r}'(t_0)$ and the curve?

1. $\mathbf{r}(t) = t^2\mathbf{i} + t\mathbf{j}$, $t_0 = 2$
2. $\mathbf{r}(t) = (1 + t)\mathbf{i} + t^3\mathbf{j}$, $t_0 = 1$
3. $\mathbf{r}(t) = \cos t\mathbf{i} + \sin t\mathbf{j}$, $t_0 = \frac{\pi}{2}$
4. $\mathbf{r}(t) = 3\sin t\mathbf{i} + 4\cos t\mathbf{j}$, $t_0 = \frac{\pi}{2}$
5. $\mathbf{r}(t) = \langle e^t, e^{2t}\rangle$, $t_0 = 0$
6. $\mathbf{r}(t) = \langle e^{-t}, e^t\rangle$, $t_0 = 0$

Differentiation of Vector-Valued Functions In Exercises

Finding Intervals on Which a Curve Is Smooth In Exercises 29–38, find the open interval(s) on which the curve given by the vector-valued function is smooth.

29. $\mathbf{r}(t) = t^2\mathbf{i} + t^3\mathbf{j}$
30. $\mathbf{r}(t) = \dfrac{1}{t-1}\mathbf{i} + 3t\mathbf{j}$
31. $\mathbf{r}(\theta) = 2\cos^3\theta\mathbf{i} + 3\sin^3\theta\mathbf{j}$
32. $\mathbf{r}(\theta) = (\theta + \sin\theta)\mathbf{i} + (1 - \cos\theta)\mathbf{j}$
33. $\mathbf{r}(\theta) = (\theta - 2\sin\theta)\mathbf{i} + (1 - 2\cos\theta)\mathbf{j}$
34. $\mathbf{r}(t) = \dfrac{2t}{8+t^3}\mathbf{i} + \dfrac{2t^2}{8+t^3}\mathbf{j}$
35. $\mathbf{r}(t) = (t-1)\mathbf{i} + \dfrac{1}{t}\mathbf{j} - t^2\mathbf{k}$
36. $\mathbf{r}(t) = e^t\mathbf{i} - e^{-t}\mathbf{j} + 3t\mathbf{k}$
37. $\mathbf{r}(t) = t\mathbf{i} - 3t\mathbf{j} + \tan t\mathbf{k}$
38. $\mathbf{r}(t) = \sqrt{t}\,\mathbf{i} + (t^2 - 1)\mathbf{j} + \frac{1}{4}t\mathbf{k}$

Using Properties of the Derivative In Exercises 39 and 40, use the properties of the derivative to find the following.

7 and 8, find $\mathbf{r}'(t)$, $\mathbf{r}(t_0)$, and $\mathbf{r}'(t_0)$ for the given value of t_0. Then sketch the space curve represented by the vector-valued function, and sketch the vectors $\mathbf{r}(t_0)$ and $\mathbf{r}'(t_0)$.

7. $\mathbf{r}(t) = 2\cos t\mathbf{i} + 2\sin t\mathbf{j} + t\mathbf{k}$, $t_0 = \dfrac{3\pi}{2}$

8. $\mathbf{r}(t) = t\mathbf{i} + t^2\mathbf{j} + \frac{3}{2}\mathbf{k}$, $t_0 = 2$

Finding a Derivative **In Exercises 9–20, find $\mathbf{r}'(t)$.**

9. $\mathbf{r}(t) = t^3\mathbf{i} - 3t\mathbf{j}$

10. $\mathbf{r}(t) = \sqrt{t}\,\mathbf{i} + (1 - t^3)\mathbf{j}$

11. $\mathbf{r}(t) = \langle 2\cos t, 5\sin t\rangle$

12. $\mathbf{r}(t) = \langle t\cos t, -2\sin t\rangle$

13. $\mathbf{r}(t) = 6t\mathbf{i} - 7t^2\mathbf{j} + t^3\mathbf{k}$

14. $\mathbf{r}(t) = \dfrac{1}{t}\mathbf{i} + 16t\mathbf{j} + \dfrac{t^2}{2}\mathbf{k}$

15. $\mathbf{r}(t) = a\cos^3 t\mathbf{i} + a\sin^3 t\mathbf{j} + \mathbf{k}$

16. $\mathbf{r}(t) = 4\sqrt{t}\,\mathbf{i} + t^2\sqrt{t}\,\mathbf{j} + \ln t^2\mathbf{k}$

17. $\mathbf{r}(t) = e^{-t}\mathbf{i} + 4\mathbf{j} + 5te^t\mathbf{k}$

18. $\mathbf{r}(t) = \langle t^3, \cos 3t, \sin 3t\rangle$

19. $\mathbf{r}(t) = \langle t\sin t, t\cos t, t\rangle$

20. $\mathbf{r}(t) = \langle \arcsin t, \arccos t, 0\rangle$

Higher-Order Differentiation **In Exercises 21–24, find (a) $\mathbf{r}'(t)$, (b) $\mathbf{r}''(t)$, and (c) $\mathbf{r}'(t) \cdot \mathbf{r}''(t)$.**

21. $\mathbf{r}(t) = t^3\mathbf{i} + \frac{1}{2}t^2\mathbf{j}$

22. $\mathbf{r}(t) = (t^2 + t)\mathbf{i} + (t^2 - t)\mathbf{j}$

23. $\mathbf{r}(t) = 4\cos t\mathbf{i} + 4\sin t\mathbf{j}$

24. $\mathbf{r}(t) = 8\cos t\mathbf{i} + 3\sin t\mathbf{j}$

Higher-Order Differentiation **In Exercises 25–28, find (a) $\mathbf{r}'(t)$, (b) $\mathbf{r}''(t)$, (c) $\mathbf{r}'(t) \cdot \mathbf{r}''(t)$, and (d) $\mathbf{r}'(t) \times \mathbf{r}''(t)$.**

25. $\mathbf{r}(t) = \frac{1}{2}t^2\mathbf{i} - t\mathbf{j} + \frac{1}{6}t^3\mathbf{k}$

26. $\mathbf{r}(t) = t^3\mathbf{i} + (2t^2 + 3)\mathbf{j} + (3t - 5)\mathbf{k}$

27. $\mathbf{r}(t) = \langle \cos t + t\sin t, \sin t - t\cos t, t\rangle$

28. $\mathbf{r}(t) = \langle e^{-t}, t^2, \tan t\rangle$

(a) $\mathbf{r}'(t)$ (b) $\dfrac{d}{dt}[3\mathbf{r}(t) - \mathbf{u}(t)]$ (c) $\dfrac{d}{dt}(5t)\mathbf{u}(t)$

(d) $\dfrac{d}{dt}[\mathbf{r}(t) \cdot \mathbf{u}(t)]$ (e) $\dfrac{d}{dt}[\mathbf{r}(t) \times \mathbf{u}(t)]$ (f) $\dfrac{d}{dt}\mathbf{r}(2t)$

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39. $\mathbf{r}(t) = t\mathbf{i} + 3t\mathbf{j} + t^2\mathbf{k}$, $\mathbf{u}(t) = 4t\mathbf{i} + t^2\mathbf{j} + t^3\mathbf{k}$

40. $\mathbf{r}(t) = t\mathbf{i} + 2\sin t\mathbf{j} + 2\cos t\mathbf{k}$

$\mathbf{u}(t) = \dfrac{1}{t}\mathbf{i} + 2\sin t\mathbf{j} + 2\cos t\mathbf{k}$

Using Two Methods **In Exercises 41 and 42, find (a) $\dfrac{d}{dt}[\mathbf{r}(t) \cdot \mathbf{u}(t)]$ and (b) $\dfrac{d}{dt}[\mathbf{r}(t) \times \mathbf{u}(t)]$ in two different ways.**

(i) Find the product first, then differentiate.

(ii) Apply the properties of Theorem 12.2.

41. $\mathbf{r}(t) = t\mathbf{i} + 2t^2\mathbf{j} + t^3\mathbf{k}$, $\mathbf{u}(t) = t^4\mathbf{k}$

42. $\mathbf{r}(t) = \cos t\mathbf{i} + \sin t\mathbf{j} + t\mathbf{k}$, $\mathbf{u}(t) = \mathbf{j} + t\mathbf{k}$

Finding an Indefinite Integral **In Exercises 43–50, find the indefinite integral.**

43. $\displaystyle\int (2t\mathbf{i} + \mathbf{j} + \mathbf{k})\,dt$

44. $\displaystyle\int \left(4t^3\mathbf{i} + 6t\mathbf{j} - 4\sqrt{t}\,\mathbf{k}\right)dt$

45. $\displaystyle\int \left(\frac{1}{t}\mathbf{i} + \mathbf{j} - t^{3/2}\mathbf{k}\right)dt$

46. $\displaystyle\int \left(\ln t\mathbf{i} + \frac{1}{t}\mathbf{j} + \mathbf{k}\right)dt$

47. $\displaystyle\int \left[(2t - 1)\mathbf{i} + 4t^3\mathbf{j} + 3\sqrt{t}\,\mathbf{k}\right]dt$

48. $\displaystyle\int (e^t\mathbf{i} + \sin t\mathbf{j} + \cos t\mathbf{k})\,dt$

49. $\displaystyle\int \left(\sec^2 t\mathbf{i} + \frac{1}{1 + t^2}\mathbf{j}\right)dt$

50. $\displaystyle\int (e^{-t}\sin t\mathbf{i} + e^{-t}\cos t\mathbf{j})\,dt$

Evaluating a Definite Integral **In Exercises 51–56, evaluate the definite integral.**

51. $\displaystyle\int_0^1 (8t\mathbf{i} + t\mathbf{j} - \mathbf{k})\,dt$

52. $\displaystyle\int_{-1}^{1} \left(t\mathbf{i} + t^3\mathbf{j} + \sqrt[3]{t}\,\mathbf{k}\right)dt$

53. $\displaystyle\int_0^{\pi/2} [(a\cos t)\mathbf{i} + (a\sin t)\mathbf{j} + \mathbf{k}]\,dt$

54. $\displaystyle\int_0^{\pi/4} [(\sec t\tan t)\mathbf{i} + (\tan t)\mathbf{j} + (2\sin t\cos t)\mathbf{k}]\,dt$

55. $\displaystyle\int_0^2 (t\mathbf{i} + e^t\mathbf{j} - te^t\mathbf{k})\,dt$

56. $\displaystyle\int_0^3 \|t\mathbf{i} + t^2\mathbf{j}\|\,dt$

Finding an Antiderivative **In Exercises 57–62, find $\mathbf{r}(t)$ that satisifies the initial condition(s).**

57. $\mathbf{r}'(t) = 4e^{2t}\mathbf{i} + 3e^t\mathbf{j}$, $\mathbf{r}(0) = 2\mathbf{i}$

58. $\mathbf{r}'(t) = 3t^2\mathbf{j} + 6\sqrt{t}\,\mathbf{k}$, $\mathbf{r}(0) = \mathbf{i} + 2\mathbf{j}$

59. $\mathbf{r}''(t) = -32\mathbf{j}$, $\mathbf{r}'(0) = 600\sqrt{3}\mathbf{i} + 600\mathbf{j}$, $\mathbf{r}(0) = \mathbf{0}$

60. $\mathbf{r}''(t) = -4\cos t\mathbf{j} - 3\sin t\mathbf{k}$, $\mathbf{r}'(0) = 3\mathbf{k}$, $\mathbf{r}(0) = 4\mathbf{j}$

61. $\mathbf{r}'(t) = te^{-t^2}\mathbf{i} - e^{-t}\mathbf{j} + \mathbf{k}$, $\mathbf{r}(0) = \frac{1}{2}\mathbf{i} - \mathbf{j} + \mathbf{k}$

62. $\mathbf{r}'(t) = \dfrac{1}{1 + t^2}\mathbf{i} + \dfrac{1}{t^2}\mathbf{j} + \dfrac{1}{t}\mathbf{k}$, $\mathbf{r}(1) = 2\mathbf{i}$

73. $\dfrac{d}{dt}\{\mathbf{r}(t) \cdot [\mathbf{u}(t) \times \mathbf{v}(t)]\} = \mathbf{r}'(t) \cdot [\mathbf{u}(t) \times \mathbf{v}(t)] + \mathbf{r}(t) \cdot [\mathbf{u}'(t) \times \mathbf{v}(t)] + \mathbf{r}(t) \cdot [\mathbf{u}(t) \times \mathbf{v}'(t)]$

74. If $\mathbf{r}(t) \cdot \mathbf{r}(t)$ is a constant, then $\mathbf{r}(t) \cdot \mathbf{r}'(t) = 0$.

75. **Particle Motion** A particle moves in the xy-plane along the curve represented by the vector-valued function $\mathbf{r}(t) = (t - \sin t)\mathbf{i} + (1 - \cos t)\mathbf{j}$.

(a) Use a graphing utility to graph $\mathbf{r}$. Describe the curve.

(b) Find the minimum and maximum values of $\|\mathbf{r}'\|$ and $\|\mathbf{r}''\|$.

76. **Particle Motion** A particle moves in the yz-plane along the curve represented by the vector-valued function $\mathbf{r}(t) = (2\cos t)\mathbf{j} + (3\sin t)\mathbf{k}$.

(a) Describe the curve.

(b) Find the minimum and maximum values of $\|\mathbf{r}'\|$ and $\|\mathbf{r}''\|$.

77. **Perpendicular Vectors** Consider the vector-valued function $\mathbf{r}(t) = (e^t\sin t)\mathbf{i} + (e^t\cos t)\mathbf{j}$. Show that $\mathbf{r}(t)$ and $\mathbf{r}'(t)$ are always perpendicular to each other.

78. **HOW DO YOU SEE IT?** The graph shows a vector-valued function $\mathbf{r}(t)$ for $0 \le t \le 2\pi$ and its derivative $\mathbf{r}'(t)$ for several values of t.

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WRITING ABOUT CONCEPTS

63. Differentiation State the definition of the derivative of a vector-valued function. Describe how to find the derivative of a vector-valued function and give its geometric interpretation.

64. Integration How do you find the integral of a vector-valued function?

65. Using a Derivative The three components of the derivative of the vector-valued function **u** are positive at $t = t_0$. Describe the behavior of **u** at $t = t_0$.

66. Using a Derivative The z-component of the derivative of the vector-valued function **u** is 0 for t in the domain of the function. What does this imply about the graph of **u**?

Proof **In Exercises 67–74, prove the property. In each case, assume r, u, and v are differentiable vector-valued functions of t in space, w is a differentiable real-valued function of t, and c is a scalar.**

67. $\frac{d}{dt}[c\mathbf{r}(t)] = c\mathbf{r}'(t)$

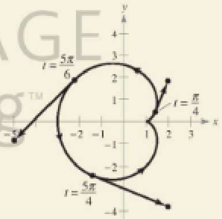
68. $\frac{d}{dt}[\mathbf{r}(t) \pm \mathbf{u}(t)] = \mathbf{r}'(t) \pm \mathbf{u}'(t)$

69. $\frac{d}{dt}[w(t)\mathbf{r}(t)] = w(t)\mathbf{r}'(t) + w'(t)\mathbf{r}(t)$

70. $\frac{d}{dt}[\mathbf{r}(t) \times \mathbf{u}(t)] = \mathbf{r}(t) \times \mathbf{u}'(t) + \mathbf{r}'(t) \times \mathbf{u}(t)$

71. $\frac{d}{dt}[\mathbf{r}(w(t))] = \mathbf{r}'(w(t))w'(t)$

72. $\frac{d}{dt}[\mathbf{r}(t) \times \mathbf{r}'(t)] = \mathbf{r}(t) \times \mathbf{r}''(t)$



(a) For each derivative shown in the graph, determine whether each component is positive or negative.

(b) Is the curve smooth on the interval $[0, 2\pi]$? Explain.

True or False? **In Exercises 79–82, determine whether the statement is true or false. If it is false, explain why or give an example that shows it is false.**

79. If a particle moves along a sphere centered at the origin, then its derivative vector is always tangent to the sphere.

80. The definite integral of a vector-valued function is a real number.

81. $\frac{d}{dt}[\|\mathbf{r}(t)\|] = \|\mathbf{r}'(t)\|$

82. If **r** and **u** are differentiable vector-valued functions of t, then

$$\frac{d}{dt}[\mathbf{r}(t) \cdot \mathbf{u}(t)] = \mathbf{r}'(t) \cdot \mathbf{u}'(t).$$

13) $r'(t) = \langle 6, -14t, 3t^2 \rangle$

17) $r'(t) = \langle -e^{-t}, 0, 5e^t + 5te^t \rangle$

21) $r'(t) = \langle 3t^2, t \rangle$

$r''(t) = \langle 6t, 1 \rangle$

$r'(t) \cdot r''(t) = 18t^3 + t$

23) $r'(t) = \langle 4\sin t, 4\cos t \rangle$

$r''(t) = \langle -4\cos t, -4\sin t \rangle$

$r'(t) \cdot r''(t) = 16\sin t\cos t - 16\sin t\cos t$

25) $r'(t) = \langle t, -1, \frac{1}{2}t^2 \rangle$

$r''(t) = \langle 1, 0, t \rangle$

$r'(t) \cdot r''(t) = t - 1 + \frac{t^3}{2}$

$r'(t) \times r''(t) = \langle -t, -(t^2 - \frac{t^2}{2}), -1 \rangle$

33) $r(\theta) = (\theta - 2\sin\theta, 1 - 2\cos\theta)$

$r'(\theta) = (1 - 2\cos\theta, 0 + 2\sin\theta)$

$(-\infty, \infty)$

37) $r(t) = \langle t, -3t, \tan t \rangle$

$r'(t) = \langle 1, -3, \sec^2 t \rangle$

$t \neq \frac{\pi}{2}, \frac{3\pi}{2}$

39) $r(t) = ti + 3tj + t^2k$

$v(t) = 4ti + t^2j + t^3k$

a) $r'(t) = i + 3j + 2tk$

$u'(t) = 4i + 2tj + 3t^2k$

b) $2r'(t) - u'(t)$

$-i + (9-2t)j + (6t-3t^2)k$

c) $5t\, u'(t) + 5u(t)$

$40ti + 15t^2j + 20t^3k$

d) $r(t) \cdot u'(t) + r'(t) \cdot u(t)$

$8t + 9t^2 + 5t^4$

e) $r(t) \times u'(t) + r'(t) \times u(t)$

$8t^3i + (12t^2 - 4t^3)j + (3t^2 - 24t)k$

f) $\frac{d}{dt}[r(2t)] = 2r'(2t)$

$= 2i + 6j + 8tk$

41) a) $r(t) \cdot u(t) = t^7$

i) $7t^6$

b) $2t^6i - t^5j$

i) $12t^5i - 5t^4j$

45) $\left((\ln t)i + C_1\right) + \left(tj + C_2\right) + \left(\frac{2}{5}t^{5/2}k + C_3\right)$

47) $\left((t^2 - t)i + c_1\right) + \left(t^4 j + C_2\right) + \left(\frac{2}{3}t^{3/2}k + C_3\right)$

49) $\left((\tan t)i + C_1\right) + \left((\arctan t)j + C_2\right)$

51) $\int_0^1 (8t\, i + tj - k)\, dt$

$(4t^2)i\Big]_0^1 + \left(\frac{t^2}{2}\right)j\Big]_0^1 - tk\Big]_0^1$

$4i + \frac{1}{2}j - k$

54)

$\int_0^{\pi/4} (\sec t \tan t)i + (\tan t)j + (2\sin t \cos t)k\, dt$

$(\sec t)i\Big]_0^{\pi/4} +$

55) $2i + (e^2 - 1)j - (e^2 + 1)k$

57) $(2e^{2t})i + 3(e^t - 1)j$

61) $$\frac{\left(2 - e^{-t^2}\right)}{2} i + \left(e^{-t} - 2\right) j + (t+1) k$$